

# Adam Sayyed

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## EDUCATION

**University of Maryland, Baltimore County (UMBC) - Meyerhoff Scholar**

June 2023 - Present

B.S. Computer Science, ML / AI Specialization, GPA: 3.8

## TECHNICAL SKILLS

Languages: C++, Python, C#, Go, Java, JavaScript, Bash, Assembly, HTML/CSS, Haskell, R

Libraries/Frameworks: PyTorch, Pydantic, Pandas, NumPy, Matplotlib, FastMCP, Kubeflow, Scikit-learn, React, SciPy, NodeJS

Developer Tools: Kubernetes, Docker, AWS, GCP, Azure, Git, GitLab, CI/CD, SLURM, MPI, CUDA, OpenMP

## EXPERIENCE

**Johns Hopkins Applied Physics Laboratory, Software Engineering Intern**

June 2026 - Present

- Accelerated multi-source data ingestion pipelines by 60% through scalable Python frameworks for distributed sensor fusion, mirroring the throughput and reliability demands of high-volume backend services
- Reduced sensor data-quality failures by 70% by implementing automated validation, anomaly detection, and monitoring frameworks that improved downstream ML reliability.
- Designed a canonical schema and ETL conversion framework unifying heterogeneous sensor feeds across ML training, simulation, and analytics pipelines, eliminating data format inconsistency as a training bottleneck.

**Sandia National Laboratories, Software Engineering & Machine Learning Intern**

May 2025 - June 2026

- Developed and prototyped ML models (SVM, Random Forest, Gradient Boosting) in Python and C++ for geospatial data, boosting classification accuracy to 95% and enabling autonomous multi-object tracking for field operations
- Designed a production-grade data validation framework for spectroscopy datasets, ensuring end-to-end data integrity across scientific pipelines and identifying top predictive features for downstream modeling
- Architected LLM integration layer via Model Context Protocol (MCP), exposing automated inference APIs that eliminated manual query workflows and reduced end-to-end geospatial analysis latency

**University of Maryland Baltimore County, Undergraduate Researcher**

January 2025 - Present

- Achieved 70% stock-movement prediction accuracy by building a multimodal ML pipeline integrating FinBERT sentiment embeddings with time-series indicators across 3 live data sources (Yahoo Finance, NewsAPI, GDELT)
- Eliminated data inconsistency across 50,000 stock trading day entries by implementing temporal alignment and preprocessing modules, ensuring clean feature handoff between ingestion and model training
- Fine-tuned FinBERT and RoBERTa models on finance-specific corpora to generate domain-adapted sentiment vectors, improving embedding quality for downstream stock-movement classification

## PROJECTS

**High-Performance 3D Heat Diffusion Engine** | C++, MPI, OpenMP, CUDA, Slurm

- Developed a 3D heat diffusion simulation with CPU, multi-node, and GPU execution, implementing MPI/OpenMP hybrid parallelism and achieving near-linear scaling with about 8 times speedup on 2 nodes for a 400<sup>3</sup> grid
- Implemented CUDA acceleration and performed profiling of memory, cache, and communication bottlenecks, achieving about 1.6 times speedup on GPU versus CPU while identifying PCIe transfer limits

**UMBC DoorDash Platform** | React, Node.js, Express, MongoDB, WebSockets, REST APIs

- Engineered a full-stack campus food delivery system with React frontend, Node.js backend, MongoDB persistence, and real-time WebSocket notifications, increasing order throughput by 24% and enabling scalable transaction handling
- Implemented optimized scheduling, delivery-matching algorithms, and transaction logging, identifying peak-demand bottlenecks, optimizing worker utilization, and producing actionable operational insights for future scaling

## AWARDS

- Meyerhoff Scholar, UMBC
- University of Maryland Baltimore County Dean's List / President's List

January 2023 - Present

## CONFERENCE PRESENTATIONS

- 2025 Joint Mathematics Meetings (JMM 2025)
- IOWA Summer Undergraduate Research Conference (SURC)

January 2025

July 2024